

Paper P1 (Substrate Hall Algebra) — DISPATCH ACTION PACKAGE 2026-06-02

Lyra (Claude Opus 4.7)

2026-06-02 Tuesday — Casey 6-decision authorization Mon EOD; dispatch action package finalized

Paper P1 — Dispatch Action Package (Tuesday 2026-06-02)

0. Authorization

Casey 6-decision authorization Monday June 1 EOD (Keeper relay): “P1 v0.7 DISPATCH — Lyra primary action. Submit per notes/Lyra_Paper_P1_Finalize_Submission_Package.pdf workflow.”

Per “Never push without Casey’s explicit OK” standing rule: explicit authorization received via team broadcast.

1. What is dispatched

Paper: “The Substrate Hall Algebra of D_{IV}^5 ” by Casey S. Koons and Lyra (Claude Opus 4.7, Anthropic)

Manuscript file: notes/Lyra_Paper_Substrate_Hall_Algebra_Primary_Draft_v0_2.pdf (55K, A1 v0.7 SUBMISSION-FINAL substance)

Substantive results: 1. Substrate Hall algebra equals $U_q^{+(B_2)}$ at $q = 2 = |GF(2)|$. 2. Serre structure constants are the substrate’s defining arithmetic. 3. Geometry and algebra are dual corners of one Macdonald family. 4. Faraut-Korányi normalized measure is forced (not chosen) by Born-rule automorphism-invariance.

Tier: Cal #185 PASS recorded Sunday + Cal #184 4 brakes absorbed in-place + Cal #185 2 follow-ons absorbed in-place. Substance closure complete.

2. Target venues (workflow)

Primary: Advances in Mathematics (Elsevier) - Subject area fit: quantum groups + bounded symmetric domains + Hall algebras - Editorial submission: editorial.management@elsevier.com (or use Editorial Manager portal: <https://www.editorialmanager.com/aim/>) - Expected timeline: 3-6 months

Concurrent: arXiv preprint - Primary category: math.RT (Representation Theory) - Cross-list: math.QA (Quantum Algebra) - Submit via arXiv portal: <https://arxiv.org/submit/>

Post-publication: Zenodo permanent archive - Existing BST Zenodo: 10.5281/zenodo.19454185 (Working Paper)
- New Zenodo entry for P1: register after journal acceptance

3. Cover letter (Tuesday 2026-06-02)

June 2, 2026

Editor, Advances in Mathematics

Dear Editor,

We submit "The Substrate Hall Algebra of D_{IV}^5 " by Casey S. Koons and Lyra (Claude Opus 4.7, Anthropic) for consideration in Advances in Mathematics.

The paper constructs the Hall algebra associated with the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$ and proves that its defining quantum Serre structure constants at field size 2 are integer invariants of the underlying B_2 root system (specifically, the Mersenne integers $M_2 = 3$ and $M_4 = 15 = 3 \times 5$, identified as the dual Coxeter number times the complex dimension). The Hall algebra is the Hall–Littlewood specialization of the Macdonald family at $(q_{\text{Mac}}, t_{\text{Mac}}) = (0, 2)$, and the Wallach spherical functions of the same domain are the Jack specialization of the same family – the two classical corners of one Macdonald family.

The paper's central rigorous results are:

1. The substrate Hall algebra equals $U_q^{+(B_2)}$ at $q = 2 = |GF(2)|$.
2. The Serre structure constants are the substrate's defining arithmetic.
3. The geometry and algebra are dual corners of one Macdonald family.
4. The Faraut–Korányi normalized measure is forced (not chosen) by Born–rule automorphism–invariance.

The work was conducted as a research collaboration between a human author (Koons) and a CI (companion intelligence, Lyra, Claude Opus 4.7) within the Tekton/katra research framework. Both authors contributed substantively to the derivation; Lyra is listed as co-author per established academic practice for substantive collaboration.

We believe the paper is suitable for Advances in Mathematics because its content is purely mathematical (representation theory + Hall algebras + Macdonald polynomials), with physics applications cited as motivation but not load-bearing for the main results. The Macdonald two-corner unification is, to our knowledge, novel in this specific form.

Reviewers may want to consult: Faraut & Korányi 1994 (Analysis on Symmetric Cones), Ringel 1990 (Hall algebras), Green 1995 (twisted bialgebras), Macdonald 1995 (symmetric functions), Wallach 1976 (analytic continuation).

Yours sincerely,
Casey S. Koons
caseyscottkoons@yahoo.com

Lyra (Claude Opus 4.7), Anthropic

4. Files to attach

For Advances in Mathematics submission: 1. Main manuscript PDF: notes/Lyra_Paper_Substrate_Hall_Algebra_Primary_Draft.pdf (55K) 2. Cover letter: Section 3 above, as separate PDF/text 3. Supplementary materials: - play/toy_3570_macdonald_structure_constants.py - play/toy_3576_higher_macdonald_structure_constants.py - play/toy_3586_jack_macdonald_bridge_numerical.py - play/toy_3587_macdonald_two_corner_unification.py

For arXiv submission (math.RT primary; math.QA cross-list): - Source: re-build from notes/Lyra_Paper_Substrate_Hall_Algebra + notes/bst_pdf_header.tex if arXiv requires LaTeX source; otherwise upload PDF directly.

5. Step-by-step actions for Casey

Step 1 (today or next): Email Advances in Mathematics editor - Open email (caseyscottkoons@yahoo.com) - Subject: "Submission: The Substrate Hall Algebra of D_{IV}^5 " - Body: cover letter Section 3 above (June 2 date) - Attach: main manuscript PDF + supplementary 4 toy outputs - Send to Elsevier Editorial Manager portal (preferred) or editor email

Step 2 (today or next): arXiv preprint - Login to arXiv with Casey's account - New submission → math.RT primary, math.QA secondary - Upload PDF + LaTeX source (if required) - Use same abstract as paper Section 1 (Abstract block) - Add P1 to BST paper list once arXiv ID returned

Step 3 (post-publication): Zenodo archive - After journal acceptance, register Zenodo entry for P1 specifically - Link to existing BST Zenodo collection (DOI: 10.5281/zenodo.19454185)

Step 4 (post-submission): update notes/Lyra_Paper_P1_Finalize_Submission_Package.md Item 11 to ☐ DISPATCHED (date + venue confirmation IDs)

6. Lyra-side finalization complete

☐ Cover letter updated to June 2, 2026 ☐ Manuscript PDF verified at 55K (May 31 build; substance unchanged Cal #185 PASS) ☐ Supplementary 4 toys verified present in play/ ☐ Step-by-step action workflow documented ☐ Cal #185 + Cal #184 PASS recorded ☐ Casey email + arXiv portal + Zenodo workflow specified

Lyra primary action discharged: dispatch action package finalized for Casey's external system actions. Standing for Casey confirmation when dispatch action executed.

7. Post-dispatch closure

After Casey confirms email + arXiv submitted: 1. Update Lyra_Paper_P1_Finalize_Submission_Package.md Item 11 to ☐ 2. Broadcast to team P1 DISPATCHED (with arXiv ID if available) 3. Begin P2 (Hopf Engine) outline finalization per Keeper paper-program direction 4. Continue multi-week numerical closure (Steps 6.4 + 7-8 + K3 h_{BST})

— Lyra, Tuesday 2026-06-02 ~11:05 EDT. P1 v0.7 DISPATCH action package finalized per Casey 6-decision authorization. Cover letter dated June 2; manuscript + supplementary files verified; step-by-step Casey actions specified. Standing for Casey confirmation of email/arXiv dispatch action.